

Rafid Ahmed

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🌐 [LinkedIn](#) 🐙 [GitHub](#) 📊 [Kaggle](#)

Education

University of Dhaka

B.Sc in Computer Science & Engineering
CGPA - (3.54/4.00)

Dhaka, Bangladesh
Jan 2019 – Jan 2024

Research Interests

Natural Language Processing: Low-resource languages · Knowledge distillation · LLM reasoning & concept learning · Parameter-efficient fine-tuning.

Vision-Language Models: Medical image segmentation · OOD robustness · Multimodal adapters · Self-supervised learning · Medical VQA · Linear projection · Preference optimization.

Work Experience

AI Engineer — Penta Global Limited

Dhaka, Bangladesh
Apr 2025 – Present

Developing scalable data pipelines using AWS Lambda, S3, Glue, EC2 & SageMaker. Building intelligent AI agents and copilots powered by LLMs. Research across medical imaging, medical VQA, LLM reasoning, preference optimization, and low-resource languages.

Software Engineer — Eximus Technologies

Dhaka, Bangladesh
Mar 2024 – Feb 2025

Built and optimized web applications with Laravel, Vue.js, React.js, and MySQL. Developed REST APIs, integrated payment and payroll modules, improved UI with Blade and Tailwind CSS.

Publications

C = Conference W = Workshop

- [C] **Rafid Ahmed***, Intesar Tahmid*, Mir Sazzat Hossain*, Tasnimul Hossain Tomal, Md Fahim, Md Farhad Alam Bhuiyan. [Evaluating Large Vision Language Models on Bangla Medical Visual Question Answering: Dataset and Benchmarking](#). *ACL 2026*.
- [W] **Rafid Ahmed***, Intesar Tahmid*, Mir Sazzat Hossain*, Tasnimul Hossain Tomal, Md Fahim, Md Farhad Alam. [How Good LLMs Are at Answering Bangla Medical Visual Questions? Dataset and Benchmarking](#). *AIMedHealth Workshop at AAAI 2026*.
- [W] Md Mahir Jawad*, **Rafid Ahmed***, Ishita Sur Apan*, Tasnimul Hossain Tomal, Fabiha Haider, Mir Sazzat Hossain, Md Farhad Alam Bhuiyan. [Benchmarking Large Language Models on Bangla Dialect Translation and Dialectal Sentiment Analysis](#). *BLP Workshop at IJCNLP 2025*. **Best Paper Award**
- [W] Intesar Tahmid*, **Rafid Ahmed***, Md Mahir Jawad, Anam Borhan Uddin, Md Fahim, Md Farhad Alam Bhuiyan. [PentaML at BLP Shared Task 1: Linear Probing of Pre-trained Transformer Models for Bangla Hate Speech Detection](#). *BLP Workshop at IJCNLP 2025*.

Under Review

- [C] Tasnimul Hossain Tomal*, Intesar Tahmid*, **Rafid Ahmed***, Mir Sazzat Hossain, Md Fahim, Md Farhad Alam. [Decoupling Representation for Fine-tuning with Linear Probing and LoRA on Robust Downstream Adaptation](#).

Research Experience

Multilingual Medical VQA & VLM Optimization

Aug 2025 – Present

Collected and curated radiology images annotated by medical experts. Introduced the first multilingual English–Bangla VQA dataset in the medical domain. Research on VLM fine-tuning, preference optimization, linear projection, and prompt learning.

Published: ACL 2026 (Long Paper), AAAI 2026 (AIMedHealth)

Influence-Driven Self-Supervised Preference Optimization (IF-DPO)

Novel self-supervised DPO framework using influence functions to identify high-impact samples as synthetic rejected signals, matching standard DPO without external reward models.

Target: NeurIPS 2026  [AhmedRafid023/IF-DPO](#)

Pretrained VLM Projectors for Efficient Vision Transformer Fine-Tuning

Investigating LLaVA/Gemma projectors as lightweight re-mappers on frozen ViT backbones. Freezing ViT backbone while training a projection-head and classifier outperforms full fine-tuning.

Target: CVPR/WACV  [AhmedRafid023/ViT-image-classification](#)

Improvised LoRA Fine-tuning for Domain-Specific Tasks

Aug 2025 – Present

Empirical analysis of further pre-training and linear probing of LoRA adapters to mitigate domain-shift issues in specialized classification and generation tasks.

Submitted: LREC-COLING 2026

Technical Skills

Languages	Python, C, C++, JavaScript, Java, Kotlin, PHP
Deep Learning	PyTorch, HuggingFace, TensorFlow, TIMM
LLM / RAG	RAG, GraphRAG, Multimodal RAG, LLM Agents, Web Agents
LLM Tools	LangChain, LlamaIndex, Chroma, vLLM, Ollama, FAISS
Other	Scikit-Learn, NumPy, SciPy, Pandas, NLTK, Plotly

Honors & Awards

Awards

Best Paper Award — IJCNLP BLP Workshop [Certificate]	2025
Finalist, Robi Datathon 3.0 — Top 20 of 3,500+ teams	2024
Finalist, Robi Datathon 2.0 — Top 20 of 384+ teams	2022

Scholarships

ORCGS Doctoral Fellowship — Full tuition, stipend CRCV, University of Central Florida	Fall 2026 – 2030
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References

Md. Farhad Alam Bhuiyan Head of R&D, Penta Global Limited Co-director, CCDS, IUB Asst. Prof., CSE, Green University of Bangladesh pdcsedu@gmail.com	Md. Mahmudur Rahman Assistant Professor Department of Computer Science and Engineering University of Dhaka mahmudur@cse.du.ac.bd
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